

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 12: Ground Proximity Warning Systems (GPWS & EGPWS)

Syllabus Source: CAE Oxford (pp. 439-488) | Pass Target: >= 90% AviationExam

1. CLASSIC GPWS OPERATING MODES (MODES 1 TO 7)

Classic GPWS warns of impending Controlled Flight Into Terrain (CFIT). It relies on Radio Altimeter rate of change and aircraft configuration from 50 ft to 2,500 ft AGL:

GPWS Warning Mode	Aeronautical Trigger Condition	Cockpit Visual & Synthetic Aural Warnings
Mode 1: Excessive Sink Rate	Radio altitude sinking excessively fast.	'SINK RATE, SINK RATE' (amber); if uncorrected: 'PULL UP' (red light + continuous aural siren).
Mode 2: Terrain Closure Rate	Rapidly rising terrain beneath aircraft.	'TERRAIN, TERRAIN' (amber); if closure rate exceeds boundary: 'PULL UP, PULL UP' (red).
Mode 3: Altitude Loss After Take-Off	Barometric altitude sink after take-off or go-around before reaching 1,500 ft AGL.	'DON'T SINK, DON'T SINK' (amber voice alert repeated until positive climb established).
Mode 4: Unsafe Terrain Clearance	Landing gear or flaps NOT in landing position at low radio altitude.	Mode 4A (Gear up, speed < 190 kt): 'TOO LOW, GEAR'. Mode 4B (Flaps up, speed < 159 kt): 'TOO LOW, FLAPS'. At higher speeds: 'TOO LOW, TERRAIN'.
Mode 5: Below Glideslope	Descent more than 1.3 dots below ILS glideslope beam.	'GLIDESLOPE, GLIDESLOPE' (amber alert; volume increases as glideslope deviation deepens).
Mode 6: Bank Angle / Callouts	Excessive bank angle (> 35° to 45° depending on altitude); plus advisory altitude callouts.	'BANK ANGLE, BANK ANGLE'; plus synthetic height calls ('ONE THOUSAND', 'FIVE HUNDRED', 'MINIMUMS').
Mode 7: Windshear Detection	Reactive windshear detected via accelerometer and air data comparison.	Red WINDSHEAR light on PFD + two-tone siren + synthetic voice 'WINDSHEAR, WINDSHEAR, WINDSHEAR'.

2. ENHANCED GPWS (EGPWS / TAWS): FORWARD LOOKING TERRAIN (FLTA)

Classic GPWS had a deadly limitation: its radio altimeter looked ONLY straight down, providing zero warning when flying toward a vertical cliff or rising mountain ridge (CFIT risk):

- EGPWS / TAWS Architecture: Adds a worldwide digital terrain elevation database, obstacle database (towers, bridges), and high-precision GPS aircraft positioning.
- Forward Looking Terrain Avoidance (FLTA): Projects a virtual look-ahead caution envelope (40 to 60 seconds ahead) and warning envelope (20 to 30 seconds ahead) along the aircraft's projected flight path. Issues alerts BEFORE the aircraft arrives at rising terrain!
- Premature Descent Alert (PDA): Warns if aircraft descends significantly below the normal 3° descent profile on approach prior to reaching the runway.
- ND Terrain Display Colors: Solid Red: Terrain > 2,000 ft above aircraft; Solid Yellow: Terrain 500 ft below to 2,000 ft above; Green: Terrain 2,000 ft to 500 ft below aircraft altitude.

[!] EXAM TRAP (AVIATIONEXAM) - GPWS / EGPWS 'PULL UP' Emergency Flight Maneuver

Upon receiving any RED 'PULL UP' or 'TERRAIN AHEAD PULL UP' warning in IMC or night: - Disconnect Autopilot and Autothrottle immediately! - Apply MAXIMUM TOGA THRUST! - Roll wings level and pitch up smoothly to the PITCH LIMIT INDICATOR (PLI) or stick shaker angle! - DO NOT RETRACT GEAR OR FLAPS until terrain clearance is fully assured!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): GPWS PULL UP Escape Maneuver**

Trigger: 'PULL UP' or 'TERRAIN, PULL UP' warning. Step 1: Disconnect Autopilot and Autothrottle immediately. Step 2: Advance thrust levers to maximum TOGA thrust. Step 3: Rotate smoothly to pitch limit indicator / stick shaker margin. Step 4: Keep wings level. DO NOT change gear or flap configuration until safe climb is verified.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the primary operational advantage of an Enhanced GPWS (EGPWS / TAWS) over a conventional GPWS?**

- [A]: It measures cabin differential pressure
- [B]: It provides Forward Looking Terrain Avoidance (FLTA) using a worldwide terrain database and GPS, detecting rising terrain ahead before flying over it
- [C]: It replaces the weather radar
- [D]: It operates only in VMC

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Classic GPWS senses terrain directly downward via radio altimeter, creating blind spots against steep cliffs. EGPWS compares GPS position against a digital terrain database to warn 40-60 seconds ahead of impact.

Q2: Which GPWS mode triggers the cockpit voice warning 'DON'T SINK, DON'T SINK'?

- [A]: Mode 1: Excessive descent rate
- [B]: Mode 2: Rapid terrain closure
- [C]: Mode 3: Inadvertent altitude loss after take-off or go-around
- [D]: Mode 5: Below glideslope

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Mode 3 monitors barometric altitude loss following take-off or missed approach with gear/flaps not in landing configuration, alerting 'DON'T SINK'.

Q3: When the flight crew receives an authentic EGPWS 'PULL UP' warning in IMC, why must the landing gear be retracted?

- [A]: Gear retraction increases parasitic drag
- [B]: Retracting gear/flaps causes a momentary loss of lift (sink) and alters aircraft trim, which could result in ground impact
- [C]: Regulations prohibit gear movement
- [D]: The warning will cancel

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Standard CFIT escape procedures mandate leaving configuration untouched during initial pull-up to avoid aerodynamic settling or secondary stall while achieving maximum climb angle.